Hidden Leverage and Political Triggers: A Major Democratic Victory During Steady GDP Growth May Be a Financial Crisis Indicator

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Abstract

Twice since 1970 a Republican president's party lost its hold on Congress in a midterm held during steady growth: 1986 and 2006. Both were followed within two years by a market crash and a banking crisis. 2026 is the third such case. We ask whether the vote is informative and, if so, why the standard recession models read 2026 as safe. We build (i) a nine-factor penalised logistic model of US recession starts, 1973–2026, which beats the yield-curve benchmark out of sample (AUC 0.92 vs 0.85) and read 87% in November 2006 when trained only on earlier data, and (ii) a 17-country crisis model with the US weighted three times. Both read about 35–60% for 2006 and about 3% for October 2026. We show the low 2026 reading is largely an artefact of measurement: since 2008, risky lending has moved from supervised banks to private credit ($2.6tn), and the official non-bank credit series the models use *falls* by 6.4 points of GDP while bank loans to non-bank lenders rose from $324bn to $2.06tn. We propose a Hidden-Leverage Crisis Model (HLCM) that adds a latent hidden-leverage factor, a layered loss-transmission rule for restructured debt, a feedback multiplier for quantitative and retail flows, and a political regime shock. A simple decision rule built on the baseline model gained 12–18% from October 2007 to March 2009 while equities lost 48%.

1 Introduction

Recession and crisis models are trained on the channels through which past crises travelled. After 2008 those channels were regulated: Basel III raised bank capital, Dodd–Frank introduced stress tests, and the 2013 interagency leveraged-lending guidance discouraged banks from lending above six times earnings. Lending did not stop; it moved. US regulators acknowledged in December 2025 that the guidance had pushed activity "outside of the regulatory perimeter" [3]. Private credit funds grew from about $0.35tn in 2012 to $2.6tn at the end of 2025 [10], and banks re-entered the chain as lenders to those funds [5].

This paper makes three contributions. First, it documents a political pattern: a growth-year midterm wave against the governing party preceded the 1987 crash and savings-and-loan crisis, and the 2007–09 crisis. We argue the vote matters through three channels; a *signal* that growth is narrow, a *policy shock* to the boom sector, and a *clock* with less room to respond; each of which can be falsified. Second, it estimates two standard early-warning models and shows, with a fixed rule tested in real time, that they would have warned in 2006 but read low today because their credit inputs no longer capture where leverage sits. Third, it proposes a model that makes that hidden leverage explicit and adds two amplifiers the standard models omit: restructured, layered debt and procyclical trading by quantitative funds and retail investors.

We are careful about what the evidence can bear. Two past cases do not make a law; the title says *may*. The election enters our quantitative work only as a labelled judgement term, never as an estimated coefficient.

2 Related literature

The yield curve is the best-established recession predictor [9]. Credit booms predict banking crises across advanced economies [15, 19], and the BIS credit-to-GDP gap is the standard early-warning indicator [4]; Greenwood et al. [14] show that rapid credit and asset-price growth jointly raise crisis probability. Leverage is procyclical [1, 12], and funding and market liquidity can spiral together [6]. The 2007–09 crisis travelled through securitised, off-balance-sheet funding [13, 18]. Leveraged ETFs must trade in the direction of the market at the close [7], and automated trading can amplify short-run moves [16]. On politics, partisan cycles affect macroeconomic outcomes [2]. The Financial Stability Board's 2026 review documents the layers of leverage in private credit and calls the sector untested in a prolonged downturn [11]. We connect these strands: the credit-boom literature is right, but the credit is now booked where its standard measures cannot see it.

3 Data and case selection

3.1 Sources

Monthly US data, January 1973 to October 2026, come from FRED (Federal Reserve Bank of St. Louis): Treasury yields, Moody's Baa spread, unemployment, CPI, WTI oil, average hourly earnings (CES0500000003),

electricity CPI (CUSR0000SEHF01), bank loans to non-depository financial institutions (LNFACBM027SBOG), and BIS total and bank credit to the private non-financial sector (QUSPAM770A, QUSPBM770A). Equity prices and dividends are from Shiller; recession dates from the NBER; broker-dealer balance sheets from the Fed's Financial Accounts. The cross-country panel uses the Jordà–Schularick–Taylor Macrohistory Database [15], 17 advanced economies, with US inputs for 2017–2026 spliced from FRED and BIS. Private-credit facts are from the Federal Reserve [10], the Boston Fed [5] and the FSB [11].

3.2 Construction

Real-time discipline. Quarterly series are lagged three months so that each month uses only data published by then. **Breaks.** The bank-loan series has a reporting break in January 2025; growth rates are break-adjusted. **Inflation.** Dollar comparisons across years use CPI-U. **Non-bank credit** is total minus bank credit to the private non-financial sector, in points of GDP.

3.3 Why we isolated these cases and months

Elections. We keep midterms in which the president's party lost control while GDP was growing steadily: 1986, 2006, 2026. Waves held in recession (1974, 1982) are excluded because a vote against the governing party in a downturn carries no information the economic data lack. 2018 (a House-only change) is excluded from the focus cases because no policy push against a boom sector followed; since the election never enters an estimated coefficient, this choice does not affect any model estimate. **COVID.** March 2020–December 2021 are excluded from fitting: a pandemic is not a financial-cycle event, and including it would teach the model that recessions arrive without warning. **Validation months.** Months already in recession are dropped from validation, leaving 358 months and 30 recession starts.

4 Stylised facts

4.1 Why the election matters

Signal. A vote against the governing party during growth reveals that growth is narrow. In the year to September 2026 average hourly earnings rose 3.2% while CPI rose 3.7%, so real wages fell; electricity prices rose 4.4% (9.7% over two years) as data centres bid for power. In 2006 electricity rose 11.9%. **Policy shock.** The new majority acts against the sector carrying growth: subprime hearings followed within about four months in 2007; calls for data-centre moratoria came within two days in 2026. **Clock.** In 2006 the equity market peaked eleven months after the vote. In 2026 federal debt is 121% of GDP and government is divided, which slows any rescue. Each channel has a falsifier: broad real-wage gains, no legislative action within twelve months,

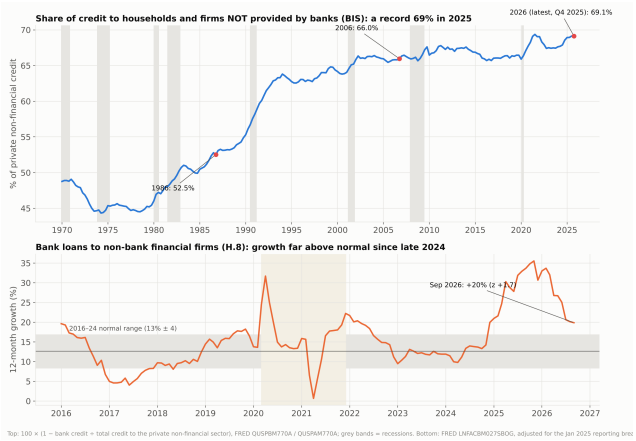


Figure 1: Share of US private non-financial credit provided outside banks.

Table 1: Debt pools, $ trillions. Pools overlap and must not be summed. AI figures are estimates.

Pool	$tn
Subprime mortgages, 2007 (2026 dollars)	2.1
Private credit funds, end-2025	2.6
Bank loans committed to non-bank lenders	2.7
of which drawn	1.7
PE-owned insurers' liabilities	0.9
Big-tech debt + off-balance-sheet AI commitments	3.0
Risky AI debt (high estimate)	0.65
AI borrowing projected, next five years	∼4

or improving credit spreads and private-credit defaults through 2027.

4.2 Where the debt went

Figure 2 summarises the relocation. Inside the perimeter, banks now lend to lenders: $1.7tn drawn and $2.7tn committed, 20% of the largest banks' loans against 9% in 2020, with the top five banks holding 63% of lending commitments to business development companies (BDCs) [5]. Outside it, private credit funds hold $2.6tn [10]; private-equity-owned insurers carry about $900bn of liabilities (from $67bn in 2012); private-credit CLOs total $155bn; and 34% of 2025 private-credit deals were AI-related against a five-year average of 17% [11]. Fund shares are held by pension funds (42%), households (23%) and life insurers (14%). Stress appears first where marking is least frequent: Tricolor and First Brands failed in September 2025, large funds limited withdrawals in February and June 2026, and the private-credit default rate reached a record 6.3%. Figure 1 shows the non-bank share of private credit rising to a record 69%.

4.3 Valuations are not the channel

A standard objection is that AI-hardware forward P/E ratios (mostly 5–30×) are far below the Nasdaq-100's roughly 60× of March 2000. This is correct and argues against a dot-com-style valuation crash. It does not argue against a credit crisis: in June 2006 homebuilders

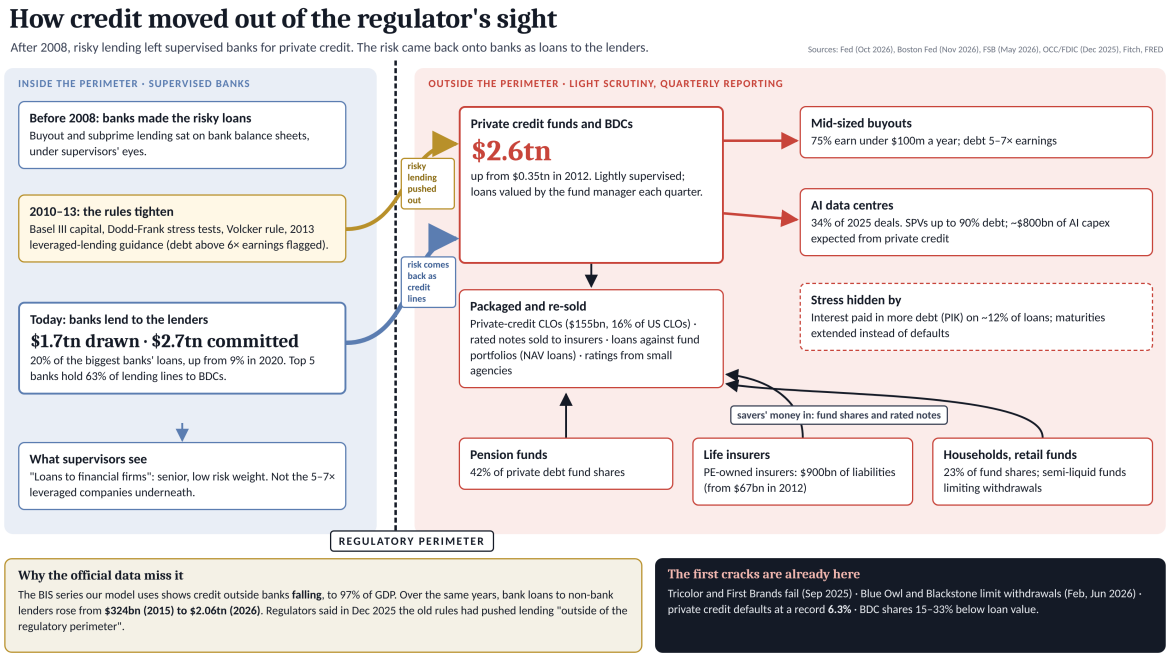


Figure 2: How credit moved out of the regulator's sight. Risky lending left supervised banks after the 2010–13 rules; the risk returned to banks as loans to non-bank lenders, which supervisors see as senior loans to financial firms. Sources as in text.

traded at 4–5× trailing earnings because profits were at a cyclical peak; the sector then fell more than 70%. Memory-chip revenue in 2026 likewise rose on price, not volume. The analogy to housing is a *necessity good priced for scarcity*: everyone needs a home, every firm will need AI, and in both cases the belief was that prices only rise. The crack in 2026 is that the price of what AI sells is falling: Chinese open models list at 60–90% below US frontier models per token.

5 Baseline models

5.1 Model 1: US recession probability

Let $Y_t = 1$ if an NBER recession starts within the next twelve months. With standardised factors $\tilde{x}_{it} = (x_{it} - \bar{x}_i)/s_i$,

$$\Pr(Y_t = 1 \mid x_t) = \Lambda\Big(\beta_0 + \sum_{i=1}^{9} \beta_i \tilde{x}_{it}\Big), \quad \Lambda(z) = \frac{1}{1+e^{-z}}. \tag{1}$$

Coefficients minimise the negative log-likelihood plus an L_2 penalty $\frac{1}{2C}\|\beta\|^2$ with $C = 0.5$ [17].

Why this form. The outcome is binary, so the model must return a probability in $[0, 1]$; a linear probability model does not. The logit/probit family is the standard in this literature [9, 19], which makes our estimates comparable. With seven US recessions since 1973, flexible learners overfit: in the panel below a logit scored 0.77 out of sample against 0.47 for gradient boosting. The additive index also makes every reading decomposable into factor contributions.

Factor selection. We started from factors with established links to recessions and tested ten leverage measures one at a time (Table 2). The BIS credit gap scores slightly better historically but measures bank credit only; we replaced it with non-bank credit and broker-dealer leverage, trading a small loss of historical fit for coverage of where leverage now sits. That today's reading ranges from about 1% to 33% across leverage measures is itself a finding.

Table 2: Leverage measures tested (out of sample, 1990–2025).

Leverage block	AUC	Brier	Oct 2026
Credit-to-GDP gap (old model)	0.942	0.079	2.0%
No credit / leverage measure	0.880	0.115	30.6%
Shadow-credit share (level)	0.841	0.105	14.1%
Shadow-credit share, 3-yr change	0.874	0.117	30.8%
Non-bank private credit, 3-yr change	0.924	0.094	2.9%
Financial-sector debt % GDP (level)	0.807	0.130	33.3%
Financial-sector debt % GDP, 3-yr change	0.864	0.116	7.9%
Broker-dealer leverage (level)	0.811	0.126	28.8%
Broker-dealer leverage, 3-yr change	0.888	0.115	26.9%
All non-financial credit incl. government, 3-yr change	0.880	0.120	14.5%
Non-bank credit change + broker-dealer leverage change (chosen)	0.923	0.099	3.0%
Shadow share + financial-sector debt change	0.870	0.085	1.0%
Financial-sector debt change + broker-dealer leverage change	0.864	0.118	8.2%

Estimates. Table 3 reports the coefficients. The yield curve dominates. Non-bank credit growth raises risk; in October 2026 its three-year change of −6.4pp lowers the probability by about 14 points relative to its mean, the largest single effect. The gold gap carries a negative coefficient, contrary to our prior; we keep it because the research question concerns it, and report that it does not help out of sample. Figure 3 decomposes the October 2026 reading.

Validation. For each year from 1990 the model is refitted on earlier data only and used to predict that

Table 3: Model 1 coefficients (per standard deviation) and the October 2026 reading. "Effect" is the change in probability (pp) relative to the factor at its mean.

Factor	Mean	SD	β	Oct 26	Effect
Yield curve, 10y−3m (pp)	1.62	1.31	-2.42	1.05	+2.0
Baa spread, 6m change (pp)	-0.08	0.44	+0.68	-0.31	-1.2
Non-bank credit, 3y change (pp GDP)	2.79	6.30	+1.28	-6.40	-13.8
Broker-dealer leverage, 3y change	-0.37	1.84	+0.43	-0.41	-0.0
Unemployment, 12m change (pp)	-0.25	0.94	+1.12	-0.35	-0.4
WTI oil, 12m change (%)	7.56	30.19	+0.74	67.63	+2.3
S&P 500 vs 10m average (%)	4.26	6.01	-0.32	6.46	-0.4
Real house prices, 12m (%)	2.45	4.75	-0.83	-2.07	+1.6
Gold vs real-rate gap (log)	0.07	0.21	-0.26	0.01	+0.2
Intercept			-4.38		

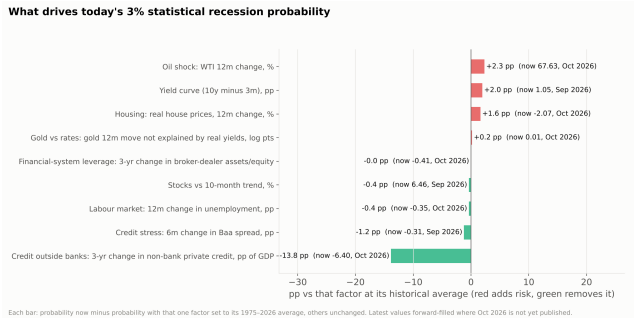


Figure 3: Contributions of each factor to the October 2026 log-odds (Model 1).

year. Table 4 compares it with a yield-curve probit tested the same way. Table 5 shows the model ranks well but is over-confident above 50%: in those months a recession followed 35% of the time. Low readings are reliable.

Readings. In November 2006 the in-sample probability was 35% and the real-time probability 87%. In October 2026, $z = -3.46$ and $\Pr = 3.0\%$ (Figure 4).

5.2 Model 2: crises in 17 advanced economies

To widen the event sample we model systemic banking-crisis starts within two years in the Macrohistory panel: 609 country-years, 1975–2014, 23 crises (US: 1984, 2007) and 46 positive rows. Ten factors include two- and five-year credit growth, real house and equity prices, the yield curve, the current account, public debt, inflation, GDP growth and global credit growth. Table 6 shows the cost of weighting the US by its share of GDP: with only two US crises, the effective sample shrinks and year-by-year accuracy falls to 0.47. Tripling the US weight is our compromise. The models compared in Table 7 confirm that the simple logit generalises best forward in time. The panel reads about 60% for the US in 2006 and about 3% in 2026, driven mainly by global credit growth (+8.6pp then, −2.3pp now). Figures 5 and 6 show the weighting comparison and the US path.

Table 4: Out-of-sample performance, 1990–2025 (358 months, 30 events).

Model	AUC	Brier
Full model (9 factors)	0.923	0.099
Yield-curve probit (benchmark)	0.853	0.122
without yield curve	0.689	0.123
without non-bank credit	0.888	0.115
without broker-dealer leverage	0.924	0.094
without gold gap	0.931	0.092

Table 5: Calibration of Model 1.

Predicted	Months	Mean pred.	Realised
0–10%	224	2.1%	0.9%
10–20%	28	14.6%	7.1%
20–30%	15	23.4%	13.3%
30–50%	23	39.8%	0.0%
50–100%	68	80.1%	35.3%

5.3 Can one model judge 2006 and 2026?

Partly. The comparison is fair in that a fixed rule, fitted once on fifty years, is applied to both dates and would have warned in real time. It is biased low for 2026 because a model can see only the channels it was trained on. In 2006 leverage sat with households and banks, which the inputs measure; in 2026 it sits in private credit, special-purpose vehicles and off-balance-sheet commitments, which they barely record. Both models share this blind spot, so their agreement about 2026 is weaker evidence than their agreement about 2006. We therefore treat 3% as a floor.

5.4 A decision rule

We test three rules fixed in advance (Table 8). Rule B holds equities when $\Pr < 20\%$, 50/50 equities and ten-year Treasuries between 20% and 50%, and Treasuries above 50%. Rule C exits equities only when $\Pr > 50\%$ *and* the S&P 500 is below its ten-month average. Signals act the following month and each trade costs 0.1%. The rules use revised data and Shiller monthly averages, which flatter results somewhat. Figure 7 shows the equity curves.

6 A Hidden-Leverage Crisis Model

The baseline models fail in 2026 for identifiable reasons: leverage is unmeasured, it is layered through restructured vehicles, its collateral is marked to markets in which quantitative and retail flows are procyclical, and a political regime change bears directly on its largest borrower group. We model each.

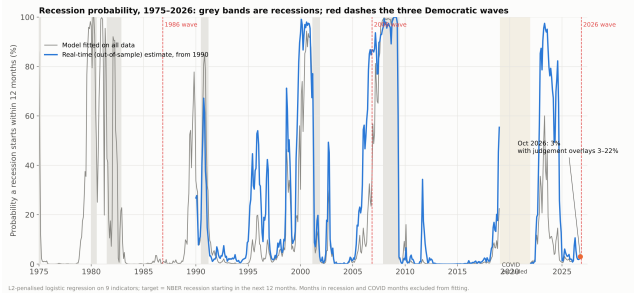


Figure 4: Model 1 probability with NBER recessions shaded.

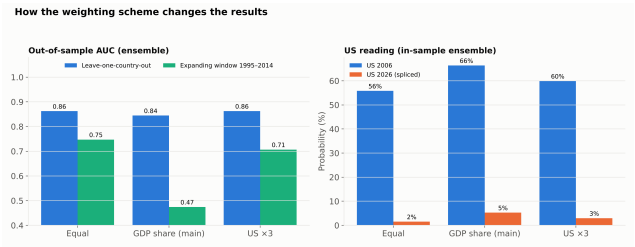


Figure 5: Accuracy and US readings under three weighting schemes.

6.1 Block A: a latent hidden-leverage factor

Let w_t be a vector of observable proxies for leverage outside the perimeter. We posit a dynamic factor model

$$w_{jt} = \lambda_j H_t + u_{jt}, \tag{2}$$
$$H_t = \rho H_{t-1} + \eta_t, \tag{3}$$

estimated by the Kalman filter with mixed frequencies [8]. Candidate proxies are bank loans to non-bank lenders, BDC price-to-NAV discounts, the private-credit default rate, the share of interest paid in kind, data-centre ABS issuance and the official non-bank credit series (with negative loading, since it falls as activity migrates). With the two series available monthly since 2015 we use the equal-weight principal component

$$H_t = \tfrac{1}{\sqrt{2}}\big[z(g_t) - z(\Delta_3 c_t)\big], \tag{4}$$

where g_t is break-adjusted growth of bank loans to non-bank lenders and $\Delta_3 c_t$ the three-year change in official non-bank credit. In September 2026 $g_t = 19.9\%$, $\Delta_3 c_t = -6.4$pp and $H_t = 0.93$ (Figure 8). The two inputs are negatively correlated (-0.20): the measure the baseline model uses moves opposite to the measure that tracks the migration.

6.2 Block B: layered loss transmission

Restructured debt places several leveraged claimants between an asset and the senior lender. Consider an asset worth 1 financed with debt share d_1; the holder of that debt finances it with debt share d_2; and so on for K layers.

Definition 1. The *cumulative senior share* of layer K is $D_K = \prod_{i=1}^{K} d_i$, the fraction of the asset's value owed ultimately to layer-K lenders.

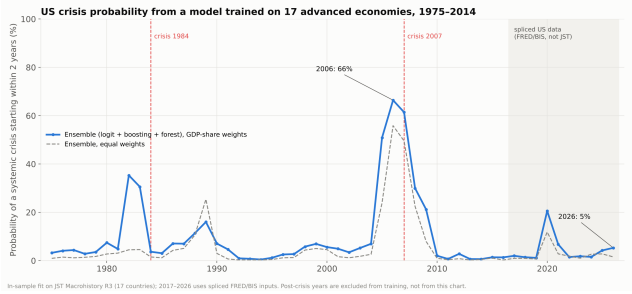


Figure 6: Model 2: probability of a US crisis within two years (US×3 weighting).

Table 6: Panel weighting schemes (logit).

Weights	US share	LOCO	Expand.	2006	2026
Equal	6%	0.86	0.75	56%	2%
US ×3	15%	0.86	0.71	60%	3%
GDP share	40%	0.84	0.47	66%	5%

Proposition 1. *Under absolute priority, pass-through of losses and no deadweight costs, a fractional loss ℓ on the underlying asset imposes a loss on layer-K lenders if and only if $\ell > 1 - D_K$, and their fractional loss is*

$$L_K(\ell) = \frac{\max(0,\ \ell - (1 - D_K))}{D_K}. \tag{5}$$

Proof. By induction. For $K = 1$ the borrower's equity $1 - d_1$ absorbs losses first. Suppose the claim at layer $K-1$, of face D_{K-1}, loses $\max(0, \ell-(1-D_{K-1}))$. Layer $K-1$'s equity, $(1 - d_K)D_{K-1}$, absorbs this first, so layer K loses $\max(0, \ell-(1-D_{K-1})-(1-d_K)D_{K-1}) = \max(0, \ell - (1 - D_K))$ on face D_K. □

Corollary 1. *If $\ell \sim F$, the probability that the senior layer is impaired is $\Pi_K = 1 - F(1 - D_K)$. Each added leveraged layer lowers the threshold multiplicatively.*

Table 9 applies the rule to stylised chains. The parameters are assumptions consistent with reported ranges (SPVs up to 90% debt; BDC debt about 0.9 times equity; Lehman at 30.7×), not estimates. The 2006 chain was impaired by a 15% fall in collateral value; US real house prices fell 39%. A plain sponsor loan held by a BDC with a bank line needs a 72% fall to reach the bank, which explains why most private credit is not a 2008-style threat. The exception is AI data-centre finance through SPVs with back-leverage, where a 37% fall reaches the senior lender (Figure 9).

6.3 Block C: quantitative and retail amplification

Collateral for AI lending (equity stakes, GPUs, data-centre valuations) is marked to markets in which several investor types trade with the move. Let the return be a fundamental shock s_t plus price impact κ of procyclical flows $f_t = \Gamma r_t$:

$$r_t = s_t + \kappa\Gamma r_t \ \Rightarrow\ r_t = M s_t, \quad M = \frac{1}{1 - \kappa\Gamma}, \tag{6}$$

Table 7: Panel models, US×3 weighting (AUC).

Model	Leave-country-out	Year-by-year
Logistic regression	0.86	0.77
Random forest	0.86	0.70
Gradient boosting	0.86	0.47
Ensemble (mean)	0.86	0.71
Yield curve only	0.71	0.53
Credit growth only	0.70	0.56

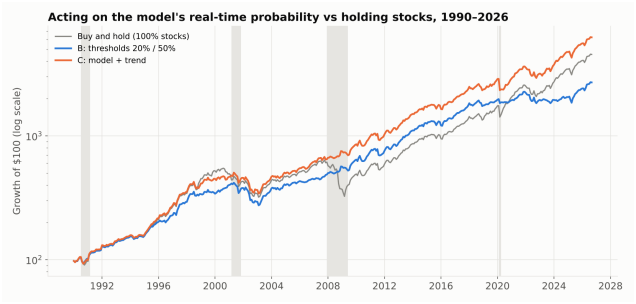


Figure 7: Growth of $100 under buy-and-hold and rules B and C, 1990–2026 (log scale); recessions shaded.

with $\Gamma = \Gamma^{\text{LETF}} + \Gamma^{\text{VT}} + \Gamma^{\text{mrg}} + \Gamma^{\text{CTA}}$. For leveraged ETFs with assets A_i and leverage L_i, end-of-day rebalancing demand is $A_i L_i (L_i - 1) r_t$ [7], so $\Gamma^{\text{LETF}} = \sum_i A_i L_i (L_i - 1)/\text{Cap}$. Volatility-targeting funds, margin-financed retail positions and trend-following funds contribute analogous terms. As $\kappa\Gamma \to 1$ the system approaches a liquidity spiral [6]. Mark-to-market losses enter Block B as $\ell^{\text{mtm}} = M\ell$. The episode of 5 November 2026, when rule-based selling of about $120bn accompanied a 4.4% fall in the S&P 500 and a 12% fall in chip stocks, is consistent with an elevated M; South Korea's retail-heavy market offers a cautionary case. We do not yet have the flow data to estimate Γ; we report scenarios.

6.4 Block D: the political regime shock

Let $D_t \in \{0, 1\}$ indicate a growth-year wave against the governing party, $R_t \in [0, 1]$ the intensity of the regulatory push on the boom sector, and E_t the boom sector's share of hidden credit (0.34 for AI in 2025). The shock $S_t = D_t R_t E_t$ shifts the expected loss on boom-sector assets, $\mu_\ell = \mu_0 + \theta S_t$, and enters the hazard directly.

6.5 The combined hazard

$$\begin{aligned}\operatorname{logit} \Pr(Y_t = 1) = \beta_0 + \beta' \tilde{x}_t + \gamma H_t \\ + \delta H_t (M_t - 1) + \psi \Pi_t + \varphi S_t,\end{aligned} \tag{7}$$

where $\Pi_t = \sum_c \omega_c \Pr(M_t \ell_c > 1 - D_c)$ is the exposure-weighted probability that senior lenders across chains c are impaired. Model 1 is the special case $\gamma = \delta = \psi = \varphi = 0$.

Link to our overlays. The judgement multipliers in our tool are a restricted version of (7). Multiplying the odds by 1.5 for private-credit stress at $H_t = 0.93$ implies $\gamma = \ln 1.5/0.93 = 0.44$; the election and regulatory multipliers (1.3×1.15) at $S_t = 0.34$ imply $\varphi = 1.18$; the leverage-and-quant multiplier of 1.15 at $M = 1.25$ implies $\delta = 0.60$. Together they move the October 2026 reading from 3.0% to 7.5%, and to 22% when all are set high.

Table 8: Backtest, Jan 1990–Sep 2026 (%).

Rule	CAGR	Vol	Sharpe	MaxDD	07–09
Buy and hold	10.9	12.3	0.69	-49	-48
60/40	8.7	7.5	0.79	-28	-26
A: scale out	10.1	9.2	0.79	-30	+14
B: thresholds	9.4	9.4	0.71	-34	+18
C: model + trend	11.9	10.8	0.85	-32	+12

Table 9: Asset fall that reaches the senior layer, $1 - D_K$ (stylised chains; parameters are assumptions).

Chain	d_i	D_K	$1 - D_K$
2006: subprime loan → SIV → investment bank	0.95, 0.93, 0.97	0.85	15%
2026: sponsor loan → BDC (bank line)	0.60, 0.47	0.28	72%
2026: sponsor loan → fund + NAV loan	0.60, 0.65	0.39	61%
2026: AI data-centre SPV → back-leverage	0.90, 0.70	0.63	37%

Scenarios. Table 10 draws the asset loss on data-centre collateral from a Beta distribution with the stated mean and concentration 8, and reports Π for two chains, without and with amplification $\kappa\Gamma = 0.4$ ($M = 1.67$). Classic BDC chains remain safe even under severe stress; levered SPV chains do not, and amplification roughly doubles their impairment probability in the stress case.

Estimation strategy. The US has no crisis since 2015, so (7) cannot be estimated on US data alone. We propose (i) estimating H_t by mixed-frequency Kalman filtering with BDC prices as the daily anchor; (ii) a Bayesian logit with priors on γ, δ, φ centred on the implied values above and on panel estimates; (iii) the sign restriction $\psi \geq 0$; and (iv) out-of-sample evaluation on the 2026–28 path, with the falsifiers stated in Section 4.

7 Results and today's signals

On the five classic signals the baseline model uses (yield-curve inversion, a 0.5pp rise in the Baa spread, a 0.5pp rise in unemployment, an oil shock, equities below trend) one fires in October 2026: oil, up 68% year on year. On four signals outside the model's data, three fire: the private-credit default rate above 6%, withdrawal limits at large funds, and bank loans to non-bank lenders growing near 20% a year; the fourth, a data-centre moratorium in a large state, is proposed but not passed. The contrast is the paper's central result: the risk is concentrated in the channels the standard data cannot see.

8 Limitations

The political pattern rests on two past cases. Model 1 predicts recessions, not crises (AUC 0.55 on crisis starts alone). Both models use revised data. The hidden-leverage series begin in 2015 and contain no

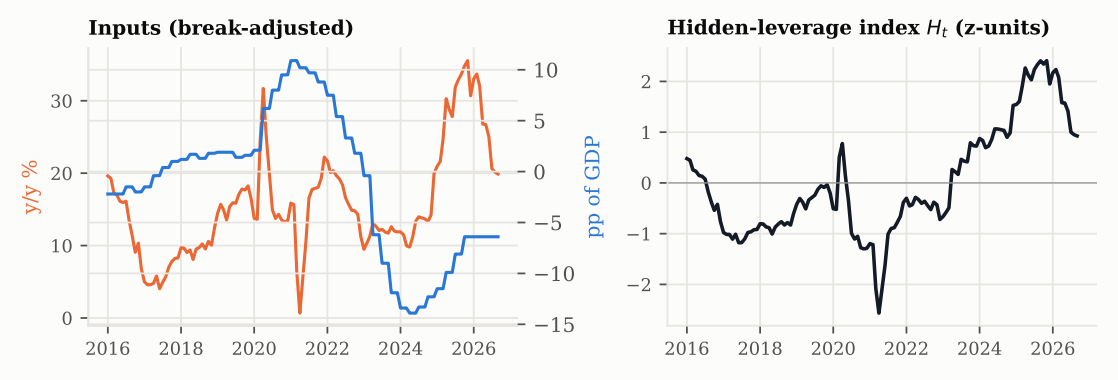


Figure 8: Inputs to the hidden-leverage index and the index H_t, 2016–2026.

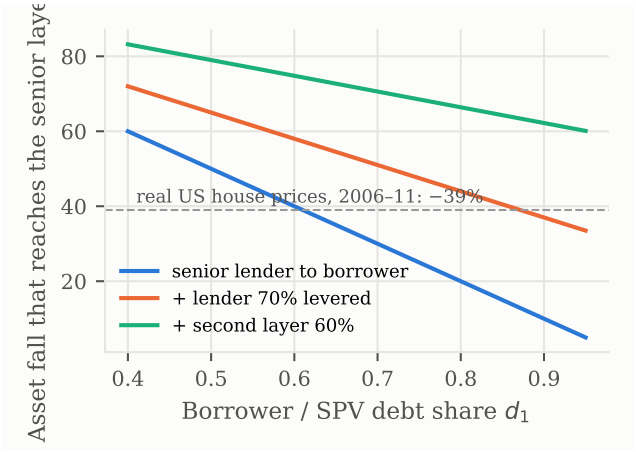


Figure 9: Threshold loss $1 - D_K$ as borrower or SPV leverage rises, for one, two and three leveraged layers.

Table 10: Probability the senior layer is impaired, Π (illustrative).

Scenario	Mean ℓ	BDC	SPV	SPV, M=1.67
Base: soft landing	10%	0.0%	2.4%	11.9%
Stress: AI demand shortfall	25%	0.3%	20.1%	51.7%
Severe: shortfall + moratoria	40%	3.1%	54.8%	85.2%

crisis. Chain parameters, loss distributions and amplification are assumptions, clearly labelled. Private credit is funded mainly by long-term capital, which limits run risk relative to 2008's overnight funding.

9 Conclusion

Standard early-warning models would have flagged 2006 and read 2026 as calm. The difference is largely measurement: risky lending left the regulated perimeter after 2008 and returned to banks as senior loans to non-bank lenders. A Democratic wave during steady growth signals that the boom is narrow, shocks the policy regime of the sector carrying it, and starts a clock. Our proposed model makes the hidden leverage explicit, shows why most private credit is resilient and where it is not; levered AI data-centre chains marked to procyclical markets; and gives testable predictions for 2027.

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